# When Better Information Makes Protests Worse: Strategic Pricing and the Collapse of Information Aggregation

**Tomoya Tajika (Nihon University)**

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@ Keio University

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# 1. Introduction

- Consider a protest demanding the withdrawal of a product
  - ex1: Nike (child labor, sweatshop)
  - ex2: Adidas (product promoting discrimination)
  - ex3: Book bans
- Question: There seems to be no obligation to respond to the protest
  - ∵ Activists are not necessarily customers
- However, if many support protests
  - The product is of little value for consumers compared with the moral issues. ↑ The firm should care
    - Products are hard to sell
    - When product sales are costly, the firm withdraws the product.

# Why is protest informative?



- If protests reveal the product's unpopularity, the sale discontinued.
  - Consumers who value the product are reluctant to support the protest.
  - Consumers who don't value it will not buy the product. They have no reluctance to participate.
- These differences in concerns make the protest's turnout informative about the product's popularity.
- This study investigates these incentives as a form of election
  - Price is key → it affects the consumers' "vote" for the product
  - Trade-off for the firm: optimal pricing vs better information
- Consider an information aggregation problem with endogenized price.

# Model



- Players: one firm and a continuum of consumers
- Timeline
  1. The firm commits to a price
  2. Consumers decide whether to participate in a protest.
  3. Players observe a public signal (= turnout + noise)
  4. The firm decides whether to withdraw the product
  5. If the product is on sale, consumers decide whether to buy it.
- Information:
  - The firm is uncertain about the product value (popularity)
  - Consumers receive noisy private signals

# Results



- Focus on the precision of individual signals received by consumers.
- Main result
  - A higher price makes protests uninformative (price–informativeness tradeoff)
  - Focus on binary signals
    - With higher signal precision, the protest's turnout becomes uninformative
    - With lower precision, an informative equilibrium exists.

# Intuition



- With higher precision, consumers receiving good signals have high WTP
  - → the firm prefers to set a high price
    - bad-signal consumers do not buy
    - good-signal consumers' purchasing benefit decreases.
    - Difference in participation motive decreases
      - → protest becomes less informative
- Independent of the timing of the price decision
  - pricing after public signal observation → a similar result holds

# 2. Model

# Model Summary



- Players: Consumers (a continuum of mass 1) and a monopolistic firm
  - Explain in the following order
    1. Consumer's utility
    2. Consumer's information (private signal)
    3. Public signal
    4. Firm's profit

# Model — Consumers



- product value: $w \cdot v$, $v \in \{0, 1\}$, each consumer has unit demand.
  - common prior: $\Pr(v = 1) = \frac{1}{2}$
  - $v$ is unknown to all players
- price: $w \cdot p$
- Cost of protest participation $c_i$, i.i.d. according to $F$,
  - allow negative values (expression motive)
- Consumer $i$'s utility ($\chi$ is an indicator)

$$u_i = [w \cdot (v - p) \cdot \chi(\text{buy})] \cdot \chi(\text{on sale}) - c_i \cdot \chi(\text{participation})$$

# Model — Information



- Each consumer receives a private signal about $v$.
  - $\theta_i \in (0,1)$ follows $G_v$
    - Assume strict MLRP (i.e., $\frac{g_1}{g_0}$ is increasing)
    - Simplify $\theta = \frac{g_1(\theta)}{g_1(\theta)+g_0(\theta)} \Rightarrow \Pr(v = 1 \mid \theta) = \theta$.

# Decision on protest participation



- Following Feddersen and Sandroni (06); Ekmekci and Lauermann (22), consumers make participation decisions to maximize consumer welfare (Rule Utilitarian)
  - Sometimes assumed in voting theory
- A justification (Ali and Lin, 13) : participate to signal being ethical
  - may be natural when "participate" includes liking or sharing on social media
  - Criticism on social media could be effective (Adidas shackle shoe on Facebook)

# Model — public signal



- All players receive public signal $t$

$$t = \text{participation ratio } (\tau) + \text{noise } (\varepsilon)$$

  - $\varepsilon \sim N(0, \sigma^2)$.

# Model — Firm's profit



- Cost
  - Continuing to sell the product costs $w \cdot K$
  - Marginal cost: $w \cdot \kappa$
- Firm's profit

$$w \cdot [(p - \kappa) \cdot \text{demand} - K] \cdot \chi(\text{on sale})$$

# Timeline



1. Firm decides the price
2. Consumers decide whether to participate
3. Players observe public signal (**signal = participation rate + noise**)
4. Firm decides whether to withdraw the product
5. If the product is on sale, consumers decide whether to buy it

# 3. Equilibrium

# After public signal



- Let $\lambda_v$ be the participation ratio when the product value is $v$
  - $\lambda = \lambda_0 - \lambda_1$
    - Higher $\lambda \iff$ public signal is more informative
    - Posteriors after observing $\theta$ and $t$ improve with higher $\lambda$ in the Blackwell sense
- Consumers' and the firm's behaviors are characterized by a threshold equilibrium
  - Consumers buy iff $\theta$ is greater than a threshold
    - threshold is increasing in $t$
  - Firm withdraws the product iff $t$ is greater than a threshold

# Consumer behavior



- Consumer buys if $\theta > \Theta_t(p) = \psi\left(2\ln(a) + \ln\left(\frac{p}{1-p}\right)\right)$, ($\psi(x) = \frac{\exp(x)}{1+\exp(x)}$)

$$\text{where} \qquad a = \frac{\Pr(v=0 \mid t)}{\Pr(v=1 \mid t)} = \exp\left(\frac{\lambda}{\sigma^2}\left[t - \frac{\lambda_0 + \lambda_1}{2}\right]\right)$$

  - $a = 1 \iff$ no information from public signal

- Demand is $1 - G_v(\Theta_v(p))$

- $\lambda_v$ : participation ratio
- $\lambda = \lambda_0 - \lambda_1$

# Withdrawal conditions



- Firm's profit is

$$\Pi(p,t) = \sum_{v\in\{0,1\}} \Pr(v \mid t)\overbrace{[1 - G_v(\Theta_t(p))]}^{\text{demand}}(p-\kappa) - K.$$

- $\kappa$: marginal cost; $K$: cost of continuing the sale
- Then, $\Pr(1 \mid t) = \frac{1}{1+a}$ and $\Pr(0 \mid t) = \frac{a}{1+a}$
  - Information wrt $t, \lambda$ is aggregated into $a$
  - Let $a^*(p)$ denote the $a$ that satisfies $\Pi(p,t) = 0$（independent of $\lambda, \sigma, t$）.
- Firm withdraws iff $t > T_f(p) \iff a > a^*(p)$

# Participation conditions



- Consider the pivotal event:

$$c_\theta(p) = \underbrace{-w \cdot [P_\theta(T_f(p)) - p]_+ \cdot \xi_\theta(T_f(p))}_{\text{marginal benefit of participation}} \geq c_i.$$

- $$\lambda_v = \int F(c_\theta(p)) g_v(\theta)\, \mathrm{d}\theta$$
$$\Rightarrow \Lambda(\lambda, p) = \lambda_0 - \lambda_1$$

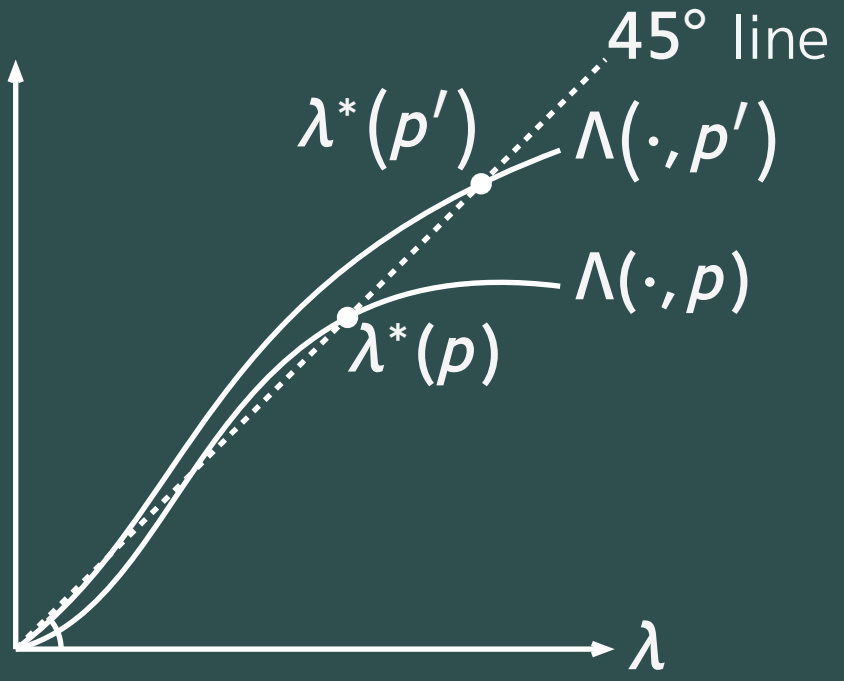
- Let $\lambda^*(p)$ be the fixed point of $\lambda = \Lambda(\lambda, p)$

$\xi_\theta(t) = \Pr(t \mid \theta)$
$[x]_+ = \max\{x, 0\}$
$P_\theta(t) = \Pr(v = 1 \mid t, \theta)$
$T_f(p)$: withdrawal threshold

# Informativeness in equilibrium



- Refer to $\lambda$ as protest **informativeness**
  - Babbling equilibrium ($\lambda = 0$) exists
  - Focus on the maximum fixed point → monotonic in $\Lambda$



# Decomposing Participation incentives



- Consumer's participation incentive

$$c_\theta(p) = -w\underbrace{[P_\theta(T_f(p)) - p]_+}_{\text{Information rent}} \cdot \underbrace{\xi_\theta(T_f(p))}_{\text{Pivotal prob.}}.$$

  - when info rent is high, less likely to participate
  - when pivotal prob is high, less likely to participate
- Intuition:
  - info-rent suppresses expression motive
  - high-WTP consumers less likely to participate
    - This makes noisy protest informative

$\xi_\theta(t) = \Pr(t \mid \theta)$

$[x]_+ = \max\{x, 0\}$

$P_\theta(t) = \Pr(v = 1 \mid t, \theta)$

$T_f(p)$: withdrawal threshold

# Decomposing Participation Incentives



- Effects on info-rent incentive

$$\frac{\partial(\text{info rent})}{\partial p} = \begin{cases} \eta_a \frac{P_\theta(1-P_\theta)}{\theta(1-\theta)} - 1 & \text{if } P_\theta > p \\ 0 & \text{if } P_\theta < p \end{cases}$$

where $\eta_a = \frac{\partial a^*}{\partial p} / a^*$ (elasticity of $a^*$)

- If $\eta_a < 0$, info-rent decreases for high $\theta$'s
  - high $\theta$'s are more likely to participate
    - protest becomes less informative
- Effects within high $\theta$'s are ambiguous



- $a = \frac{\Pr(v=0 \mid t)}{\Pr(v=1 \mid t)}$
- $a^*$: least $a$ to withdraw
- $P_\theta(t) = \Pr(v = 1 \mid t, \theta)$

# Price–Informativeness Tradeoff



## Theorem 1.

When $\sigma$ is small enough, $\frac{\partial \Lambda(\lambda,p)}{\partial p} < 0$ if

$$\lim_{\sigma \to 0} \operatorname{sgn}\left(\frac{\partial^2 c_\theta}{\partial \theta \partial p}\right) = (1 - a^*(p))\left[1 + \left[\frac{p - \frac{1}{2}}{a^*(p) - 1} + \frac{p}{2}\right]\eta_a(p)\right] > 0.$$

- $\eta_a(p)$ is the elasticity of $a^*(p)$

- Intuition: a consumer with high $\theta$ becomes more likely to participate…
  - More participation when $v = 1$
  - Informativeness: $\lambda = \lambda_0 - \lambda_1$

- $\sigma^2$: variance of public signal
- $a = \Pr(v = 0 \mid t) / \Pr(v = 1 \mid t)$
- $a^*(p)$ is $a$ when profit is 0

# Optimal price with fixed $\lambda$



- $\frac{\partial \Lambda}{\partial p}$ depends on $p$ $\Rightarrow$ consider the optimal price with fixed $\lambda$

**Definition 1.**

The optimal price with fixed $\lambda$ maximizes $\Pi^e(p;\lambda) = \mathbb{E}_t[\max\{\Pi(p,t),0\}]$.

- Ex-ante: averages over all $t$; $a^*(p)$ concerns only $t = T_f(p)$
- $p = 1$ itself is never optimal: no consumer buys

# Comparative statics around optimal price



## Proposition 1.

Suppose that $\lim_{p \to 1} \frac{1 - G_v(\Theta_t(p))}{g_v(\Theta_t(p))} = \infty$.

The optimal price with fixed $\lambda$ is $p \to 1$ and $\lim_{p \to 1} \frac{\partial \Lambda(\lambda, p)}{\partial p} < 0$ as $\sigma \to 0$.

- There is a price–informativeness tradeoff
- When $\sigma$ is high enough, the inverse relation can arise.
  - ∵ Pivotal motive is dominant

# Intuition



- $\sigma \approx 0 \Rightarrow P(v = 1 \mid t) \approx 1$ or $0$
  - $p \to 1$
  - Let $\theta(p)$ be the solution to $P_\theta(T_f(p)) = p$.
    - $\lim_{p \to 1} \theta(p) < 1$; otherwise, no positive profit
  - $a^*(p) \approx 0$ (sufficiently optimistic)
    - sale is continued iff $a \approx 0$.
  - At $a = a^*(p)$, profit is $0$
    - higher price $\to$ lower profit at $a = a^*(p)$
    - $\eta_a = -\frac{1}{a}\frac{\Pi_p(p,a)}{\Pi_a(p,a)} < 0$



- $a = \frac{\Pr(v=0 \mid t)}{\Pr(v=1 \mid t)}$
- $a^*(p)$ is $a$ when profit is 0

# Effects on info-rent



$$\frac{\partial(\text{info rent})}{\partial\theta} = \frac{P_\theta(1-P_\theta)}{\theta(1-\theta)} \quad \text{if } P_\theta > p$$

- $\eta_a < 0$
  - → higher price implies $v = 1$ is more likely at the withdrawal threshold
  - → $P_\theta$ increases → $\frac{\partial(\text{info rent})}{\partial\theta}$ decreases
  - → info rent becomes homogeneous with higher price
  - → protest becomes less informative

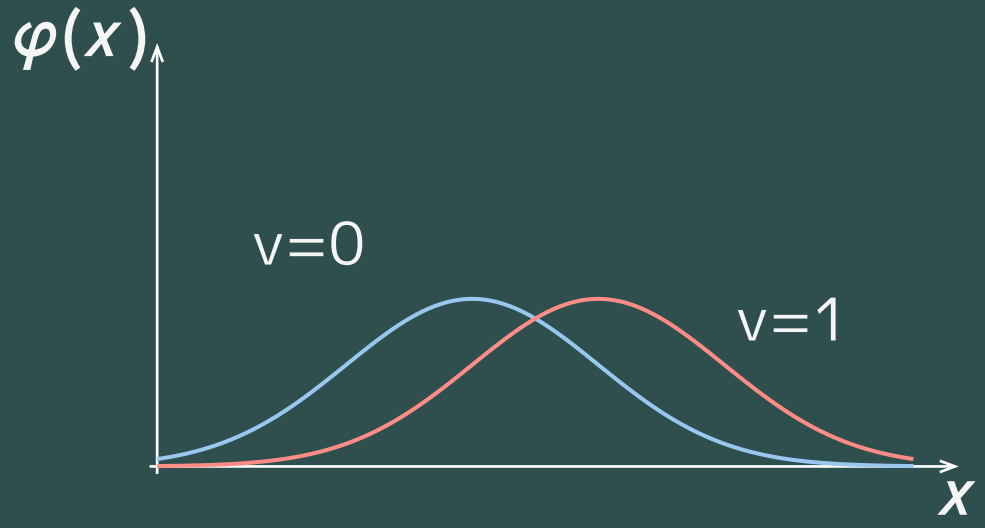
$P_\theta(t) = \Pr(v = 1 \mid t, \theta)$

# Effects on pivotal prob.



- Pivotal prob. is $\Pr(T_f \mid \theta)$
  - a mediant of $\{\Pr(T_f \mid v)\}_v$
- For smaller $a$, pivotal prob. decreases
- Higher $p \Rightarrow$ smaller $a^*$
  - compress the difference in participation motives

→ less homogeneity



- Does the firm prefer less information? → NO!

**Proposition 2.**

When $\sigma$ is small enough, profit is increasing in $\lambda$.

- Price and informativeness have a tradeoff for the firm's profit

# 4. Equilibrium under binary signal

# Binary signal



- To characterize the full equilibrium, focus on binary signals
  - Let $\theta \in \{\theta_H, \theta_L\}$
  - $\Pr(v = 1 \mid \theta = \theta_H) = \Pr(v = 0 \mid \theta = \theta_L) = \mu \in \left(\frac{1}{2}, 1\right)$
    - Refer to $\mu$ as **precision**.

# Condition for information aggregation



## Proposition 3.

Let $\overline{p} := \mu \frac{K+(1-\mu)\kappa}{\mu(1-\mu)+(2\mu-1)K}$

a. If $p \geq \overline{p}$ or $p \leq K + \kappa$: no informative equilibrium exists

b. If $K + \kappa < p < \overline{p}$, an informative equilibrium exists for sufficiently large $w$

› This result is independent of noise size

# Implication for Proposition 3



- When noise is small, $P(v = 1 \mid t) \approx 1$ or $0$
  - aggregated information is independent of precision
- Under informative equilibrium, price is lower than $\overline{p}$
  - Under some conditions, $\overline{p}$ is increasing in $\mu$
- Under the babbling equilibrium, price is lower than $\mu$
  - $\because$ highest WTP is $\mu$

→ Consumers dislike higher precision as it only raises the price

$\mu = \Pr(v = s \mid \theta = \theta_s)$

# 5. Endogenized price

# Intuition



- When precision is high
  - Large difference in WTP between H and L
  - Price is set high to exploit the H type → fail to aggregate information
- When precision is low
  - Small difference in WTP between H and L
  - Price is set low to make both types buy
    → informative equilibrium

# When signal precision is high



## Proposition 4.

Suppose that $\mu > \kappa + 4K$, $\kappa + K > \frac{1}{2}$, and $M(\mu) > 0$.

$$M(\mu) := \mu - \kappa - K\frac{\mu + \kappa(2\mu - 1)}{\mu(1-\mu) + (2\mu - 1)K} - K.$$

Then, the optimal price is $p = \mu > \overline{p}$ and no informative equilibrium exists.

- This result is independent of $\sigma$ (variance of public signal)

- $\mu$: precision of private signal
- $\kappa$: marginal cost
- $K$: cost to continue sale

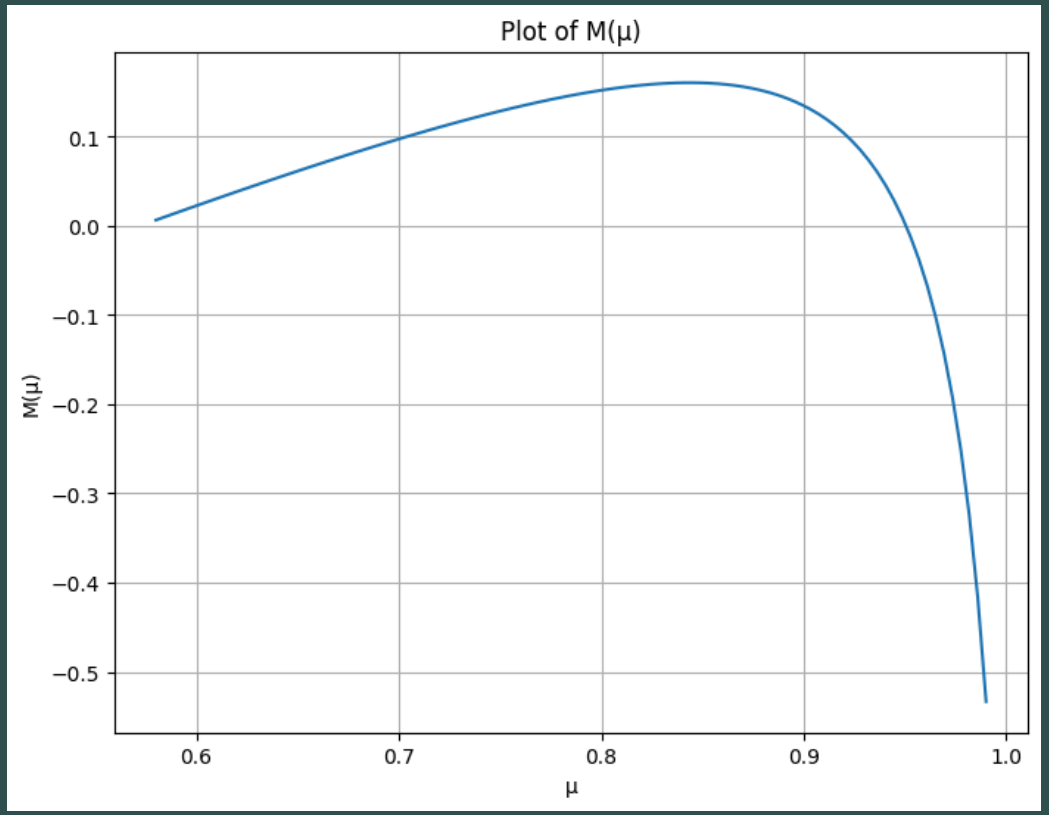



Figure 1: Graph of $M(\mu)$ when $\kappa = \frac{1}{2}$ and $K = 0.02$

# When precision is low



**Proposition 5.**

Suppose that $\mu \in (\kappa + K, \kappa + 2K)$. Then for small $\sigma$, if $w$ is large enough, there is an informative equilibrium under optimal price.

# Implication



- Informativeness is non-monotonic in precision of the private signals
  - not a common feature in protest models
  - Ekmekci and Lauermann (22) show monotonicity results
- Endogenous pricing is key

# 6. Pricing after public signal

# Pricing after public signal



- In the base model, the firm commits to a price **before** the protest
  - Consider the price decision **after** the protest
- New timeline
  1. Consumers decide whether to participate in the protest
  2. Public signal $t = \tau + \varepsilon$ is observed
  3. Firm decides whether to continue or withdraw; if it continues, it sets the price
  4. Consumers decide whether to purchase

# Difference from the previous model



- With larger $t$, price becomes lower
- This creates the incentive to increase $t$ to lower the price.
  - This incentive is stronger for the consumers who are likely to purchase.
  - H-type is more likely to purchase

→ H-type can be more likely to participate

# Results



## Proposition 6.

Under the supposition that "higher participation rate → unpopular"

a. If $\mu > \mu^{**}$ and $\frac{\mu}{2} > K$, when $\sigma$ and $w$ are large enough, $c_H > c_L$

- When $\sigma$ is small enough $c_L > c_H$

b. If $\mu^* < \mu < \mu^{**}$ and $\sigma$ and $w$ are large enough, $c_L > c_H$.

- With high precision, H-type is more incentivized to lower the price → protest fails to aggregate information

# Intuition



- If “higher participation rate → unpopular”, participation lowers the price
- With high precision, the firm sets the price to exploit H-type (L-type does not buy)
  - → H-type is more incentivized to lower the price
  - → H-type is more likely to participate
- When $\sigma$ is small, such an incentive does not work
  - If the firm believes that $v = 0$, it discontinues the product sales
    - → No benefit from such price reduction

# 7. Literature

# Related Literature



- Protest (Information aggregation, political)
  - Battaglini (17); Ekmekci and Lauermann (22)
- Boycotts
  - Discrete public goods
    - Diermeier and Van Mieghem (08); Delacote (09)
  - Signaling
    - Feddersen and Gilligan (01); Innes (06); Miyagiwa (09); Glazer et al. (11); Heijnen and van der Made (12); Peck (17)
  - War of attrition: Egorov and Harstad (17); Correa (25)
- Consumer activity: Broccardo et al. (22); Kaufmann et al. (24)

# 8. Summary

- We study consumer protests demanding the withdrawal of a product
- The firm faces a price–informativeness tradeoff: higher price implies lower informativeness.
- In a binary signal model, the tradeoff is severe: higher precision can reduce informativeness of the protest
  - This occurs whether the price is set before or after the public signal observation
- Implication: higher precision may make consumers worse off.